LS-CAT Management Board Meeting

Keith Brister
December 9, 2013

LS-CAT Management Board Meeting Agenda

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       Start Time: 9 AM CT (10 AM ET) December 9, 2013
            Audio: 1-888-850-4523
         Passcode: 190580
          Video 1: http://conf1.ls-cat.org
          Video 2: http://conf2.ls-cat.org
Physical Location: 9700 S. Cass Ave Bldg 436A, Argonne, IL
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- Approval of previous meeting's minutes
- Status
  - Alfonso Mondragón succeeds Wayne Anderson as LS-CAT Scientific Director
  - WSU drops out as of June 10, 2014
  - Detector proposal submitted
  - APS Upgrade takes a dramatic turn

- Budget update
  - Don’t panic

- LS-CAT Renewal and Rejuvenation
- APS Review of Sector 21 November 4, 2014 (this time for sure)
- Data Collection/Analysis Workshop
- New Business

Status

- Alfonso Mondragón succeeds Wayne Anderson as LS-CAT Scientific Director

- The $2M S10 proposal to the NIH for a Pilatus 3 6M detector has been submitted.
  - These grants are extremely competitive
  - We should have a priority score by the next board meeting

- WSU drops out as of June 10, 2014
  - FY14 budget (our current year) is not much affected as personnel changes just about cancel the WSU shortfall.
  - More beamtime for everyone! Or recruit new members.

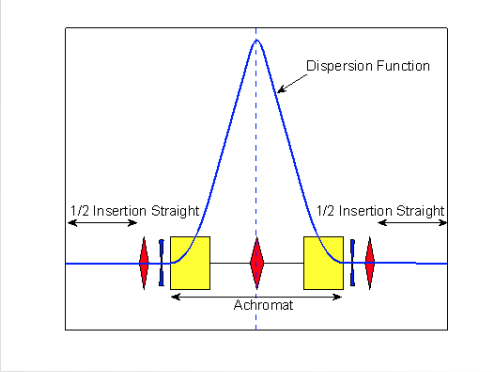
- Jason Ackley (Electromechanical Technician) left LS-CAT for the APS power supply group July 5, 2013.

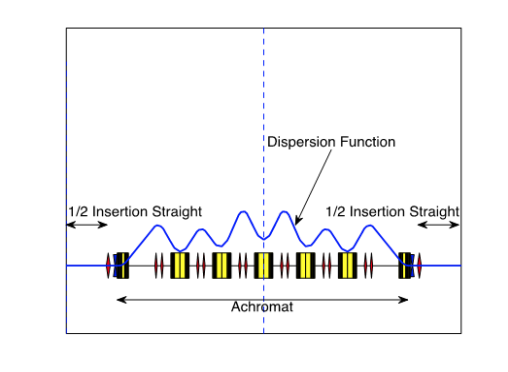
Preview of the FY15 Budget (Confidential)

Facts:

1. FY14 is 309K / share (154.5K / half share)
2. A 3% increase would put us at 318.3K / share (**Dare I propose this?**)
3. Current expenses are within the approved FY14 budget (based on the first quarter of FY14)
4. We have about 2M in our war chest
5. Moving from 6.5 shares to 5.5 shares hurts but is not devastating
6. Northwestern will likely submit a P41 to support Bionanoprobe operations.
   a. This could potentially make up for the loss of WSU
   b. Effect on beamtime would need to be negotiated with and approved by the LS-CAT Management Board
7. I intend to submit a budget of about 1.7M in May **keeping per share costs inline with what we have today**
   a. Essentially zero capital equipment (for repairs and the new detector fund)
   b. Use war chest for repairs
   c. Increased risk of staff burnout with 9 staff members instead of 10

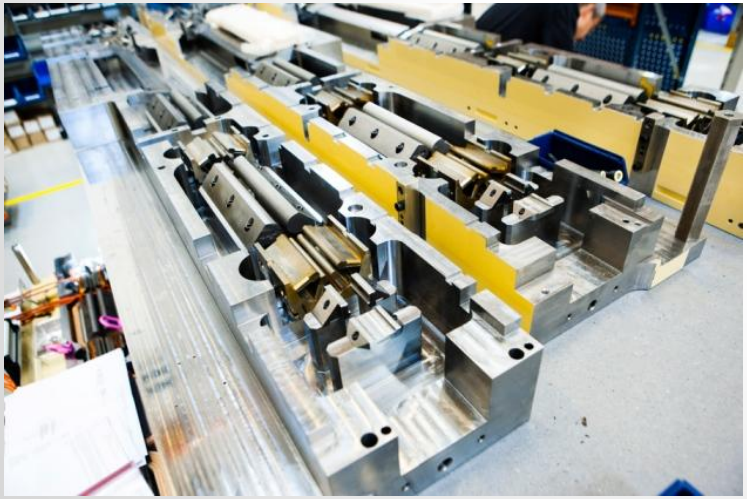
Multi Bend Achromat Lattice





http://www.aps.anl.gov/News/Meetings/Monthly_Meetings/2013/20130828_borland.pdf

Magnet assembly for Max IV



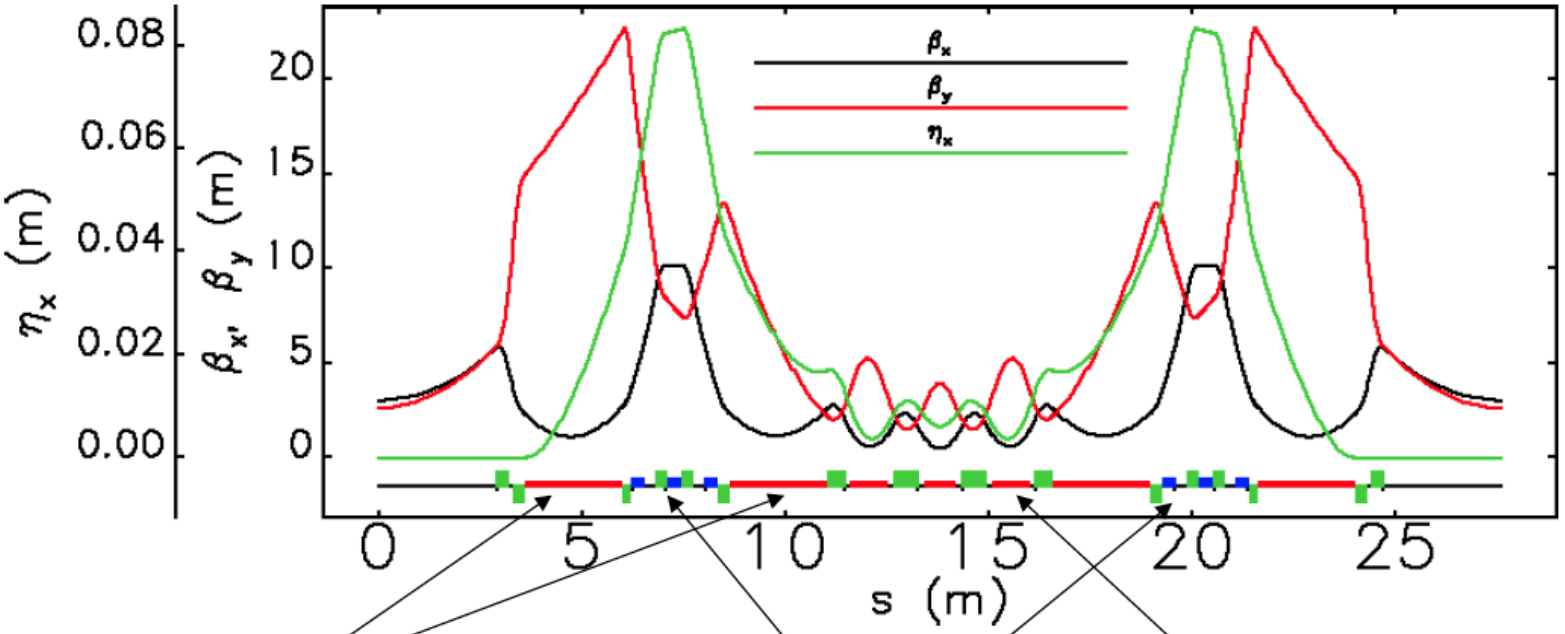
Likely facility locations



Nature 501, 148–149 (12 September 2013)

Hybrid 7 Bend Achromat APS Lattice Concept

Emittance of 80 pm or less at 6 GeV



Long dipoles with 5-segment longitudinal gradient, reduces emittance while boosting dispersion bump amplitude

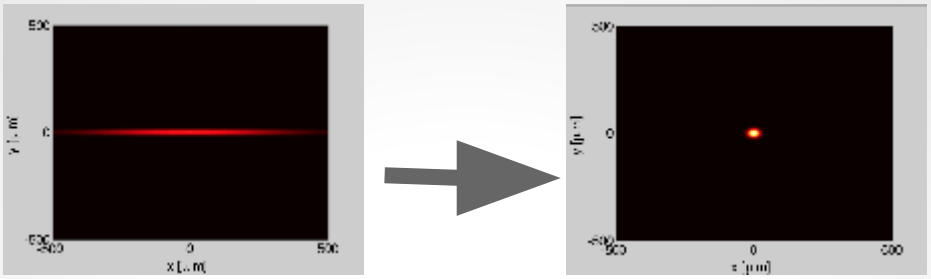
Three families of sextupoles separated by π or 3π phase advance to cancel nonlinear kicks

Transverse gradient dipoles to eliminate quads and change damping partition

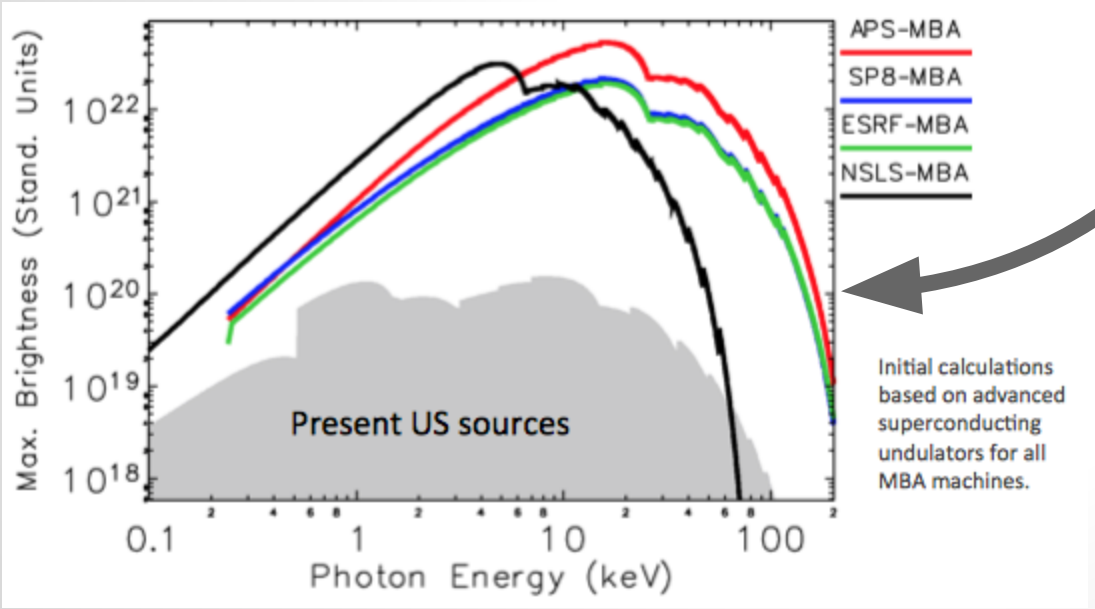
Original lattice concept from ESRF.
L. Farvacque *et al.*, Proc. IPAC13, 79 (2013).

MBA Lattice Simplified

Present storage rings produce a very wide beam.



New multi bend achromat machines will produce a round (ish) beam.

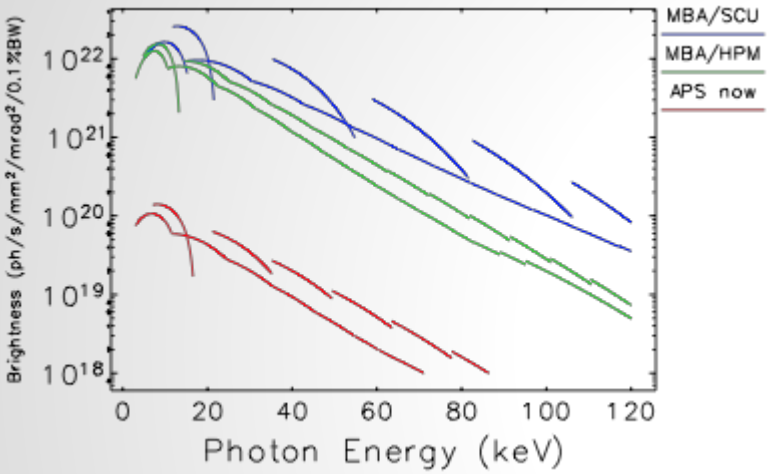


Initial calculations based on advanced superconducting undulators for all MBA machines.

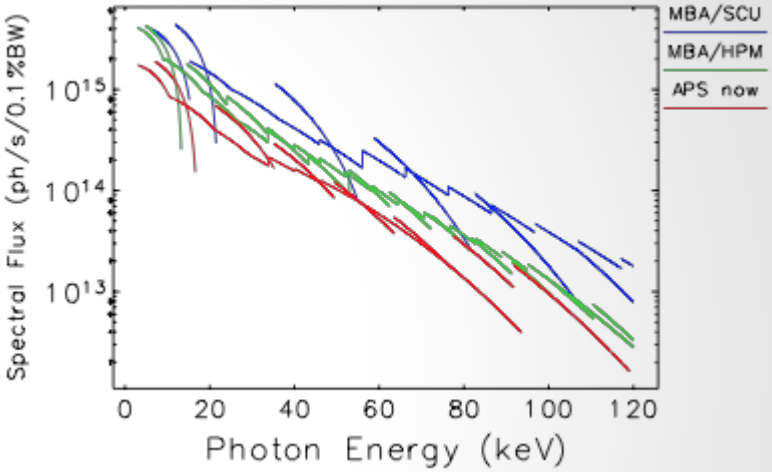
Resulting in a huge improvement in the ability to deliver a small spot of x-rays on a sample.

Likely APS MBA Upgrade Curves

Brightness



Flux



Bottom Line: If we do not upgrade, LS-CAT users will see a factor of 2 to 4 improvement. If we do upgrade they will see a factor of 100.

The APS part of the upgrade will require a 12 to 18 month down period currently thought to be around 2019

Is the MBA Lattice a good thing?

For the Bionanoprobe the answer is a resounding yes. The current BNP would be able to immediately use it with at least a factor of 10 increase in flux delivered to the sample as well as a factor 2 (or better) in spatial resolution.

A strong case for a BNP-2 could be made to further exploit the beam.

For crystallography it is also a yes but would require effort to take full advantage of.

More to the point is: What sort of crystals will you need to be working with in 2020? People are working on routine 1 micron crystals exposed once using various techniques to deliver a flux of crystals to the beam. This would require all new endstation equipment.

Understanding protein synthesis and better antibiotics

Scientific Opportunity

Ribosomes are large, complex molecular machines that translate the genetic code of mRNA to synthesize proteins. Antibiotics block protein synthesis killing bacterial infections. Polysomes (ribosome clusters) simultaneously read mRNA to synthesize proteins.

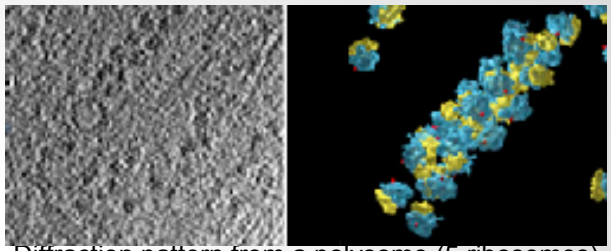
Breakthrough Techniques

Highly collimated beams were required to solve the ribosome structure. Weak scattering and extreme unit cell dimension (2700 Å) from polysome crystals require enhance microcrystallography techniques.

MBA-lattice Enabler

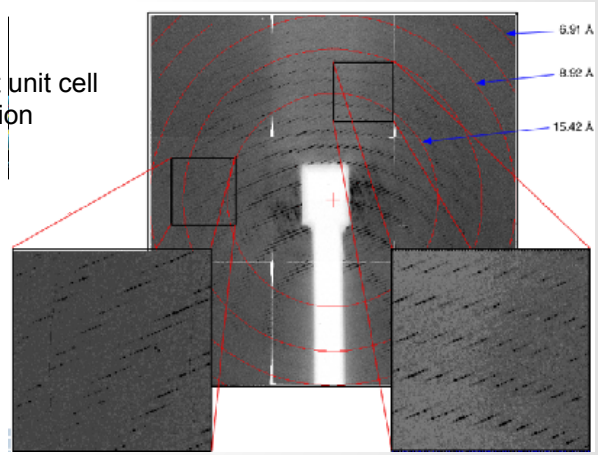
The large unit cell and weak scattering can be combated by the intense, extremely collimated beam to probe micro-crystals or well-ordered regions of larger crystals.

EM reconstruction of polysome



Diffraction pattern from a polysome (5 ribosomes)
(Courtesy of Dr. Jaime Cate, U. of California Berkeley)

Largest unit cell dimension 2700 Å



Ada Yonath shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome.



Understanding protein synthesis and better antibiotics

Scientific Opportunity

The nuclear pore complex (NPC) mediates the exchange of macromolecules between the nucleus and the cytoplasm. It is one of the largest supramolecular assemblies in the cell (120 nm in diameter, 120 Mega-Daltons, 30x larger than the ribosome). It is composed of many copies of 30 unique proteins.

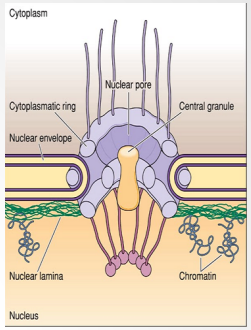
Breakthrough Techniques

Enhanced microcrystallography techniques and low noise detectors are required to solve the structure of subunits and the intact NPC.

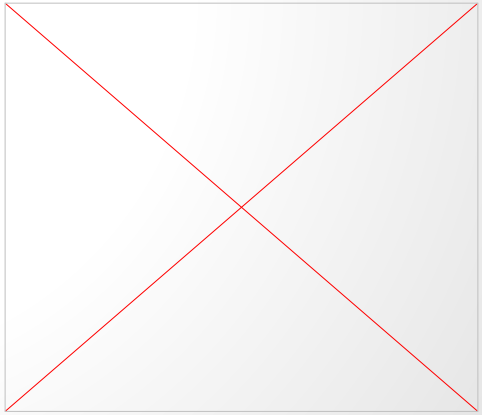
MBA-lattice Enabler

The high brightness will provide an intense, extremely collimated beam necessitated by the 120 nm dimension and to probe micro-crystals or well-ordered regions of larger crystals.

Cross section of the NPC



Multi-subunits only diffract to 9 Å



André Hoelz, California Institute of Technology

In situ Data Collection

Scientific Opportunity

In Situ screening will provide important diffraction feedback on limited quantities of biological material at an early stage in crystallization trials.

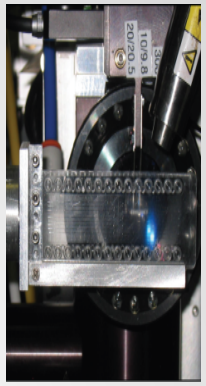
Breakthrough Techniques

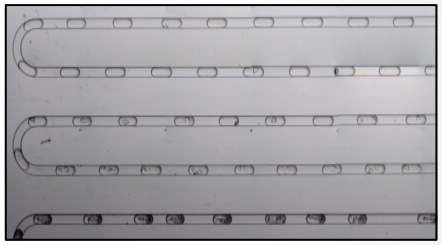
- Samples introduced by novel delivery systems (e.g. acoustic drop or microfluidics)
- Data collection on large number of micro-crystals complexed to a variety of compounds.
- Data collection on high symmetry space groups (e.g. viruses)

MBA-lattice Enabler

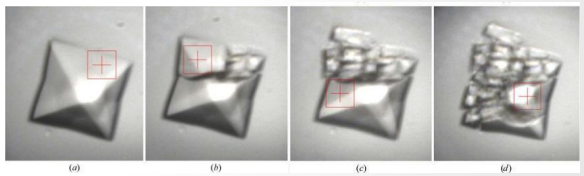
Higher brightness and faster detectors employed in the search for every shrinking crystals of increasing complexity and biological importance.

Microfluidic crystallization card on a goniometer with a mini-beam collimator





Many crystals in micro-channels for rapid data collection



Crystallization tray - containment for virus crystals. Radiation damage occurs quickly (<100 ms) requiring many crystals.

In situ data collection from a virus crystal at 3 different Positions. D. Axford et. al., Acta Cryst. (2012) D68, 592-600

Microcrystallography of Biological Macromolecules

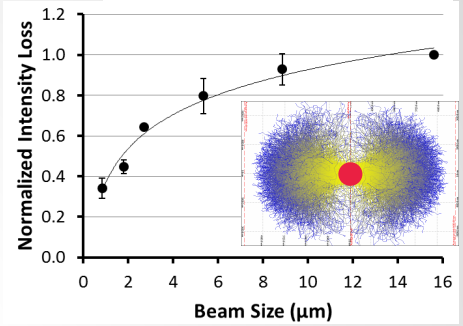
Microcrystallography has enabled the determination of high impact, 3D structures of proteins & protein complexes

APS-U & MBA-lattice will significantly expand the MX horizon

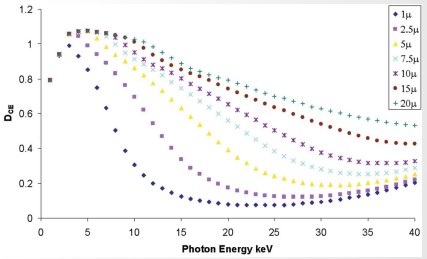
- Exploit the high Brightness
  - Nano/Microcrystals – currently inaccessible
  - Improved S/N and resolution for small (0.5 – 5 µm),
- Inhomogenous and/or weakly scattering crystals
- Exploit APS high energy source
- Reduced radiation damage – photoelectron escape
- Enhanced modes of data collection
  - *In situ*, Rastering, Helical and Shutterless data collection

Reduced Radiation Damage

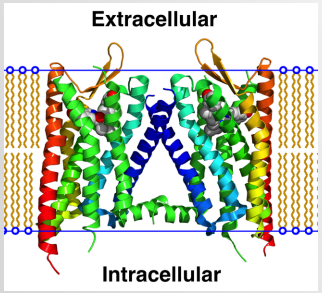
With microbeams

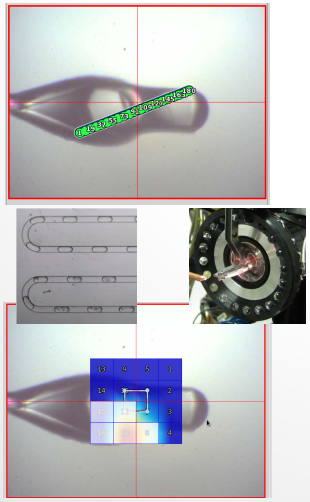


Or high energy X-rays

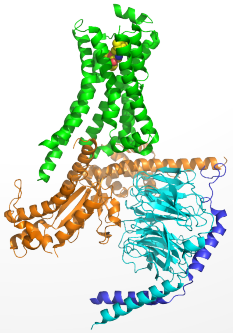


k-opioid GPCR
Ray Steven's lab





β_2 adrenergic receptor-Gs protein complex
Kobilka & Weis labs



2012 Nobel Prize in Chemistry



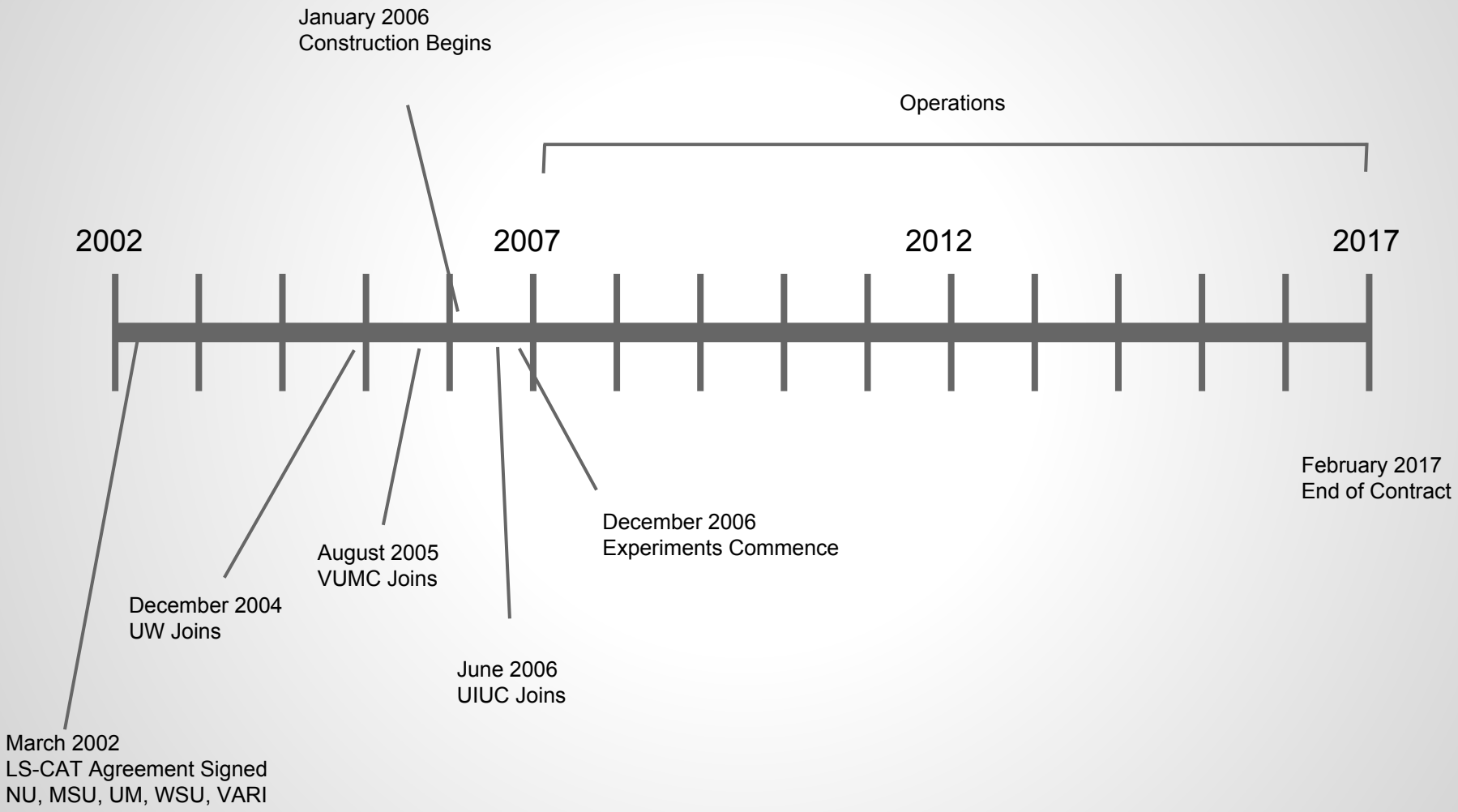
LS-CAT Extension and LS-CAT-2

Possible Approach:

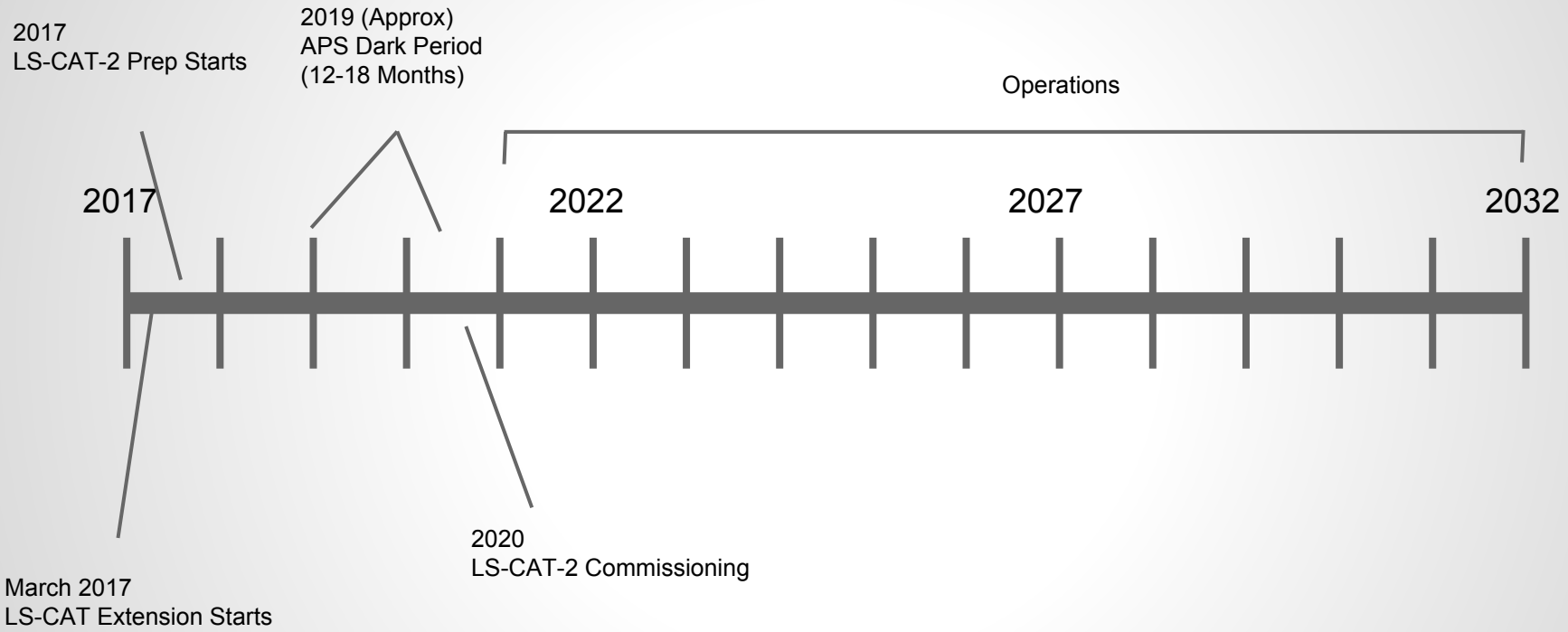
Extend, with no changes, the existing contract to run from March 2017 through the end of the APS dark period with our crystallographers possibly serving members at other synchrotron radiation sources. Practically this would be a 5 year extension with an early termination after APS-U commissioning.

Separately start a new capital project, possibly including the same partners, to rebuild LS-CAT to fully take advantage of the APS-U multi bend achromat lattice. This would be an all-new contract that would probably need 10 members to reach its full potential.

LS-CAT Timeline



LS-CAT-2 Timeline (Draft)



DRAFT LS-CAT-2 Capital Costs (if APS covers IDs)

Item	Cost Each	Cost Uncertainty	Quantity	Total	Total Uncertainty
Station Modifications	0.3M	0.2M	1	0.3M	0.2M
Detectors	2.0M	0.5M	4	8.0M	2.0M
Focusing	0.5M	0.2M	4	2.0M	0.4M
Monochromators	0.5M	0.3M	4	2.0M	1.2M
Goniometry	0.6M	0.2M	4	2.4M	0.8M
Robotics	0.4M	0.2M	4	1.6M	0.8M
BNP2	4.0M	1.0M	1	4.0M	1.0M
TOTAL				**20.3M**	**2.8M**

What an LS-CAT-2 might consist of

Undulators:

In order to keep the magnet specs for the APS-U within engineering reality the energy of the machine will need to be reduced from the current 7 GeV to, probably, 6 GeV. This means all the undulator magnetic structures will need replacement (at APS-U expense). At this point the APS-U management has expressed that fully optimized devices, including superconducting undulators, are on the table. Likely the APS life sciences community will need to persuade the NIH to help on this front.

Since the beam is round(ish) there is no reason to stick with horizontally deflecting undulators and it might be less expensive to provide vertically deflecting ones. This opens new possibilities downstream.

What an LS-CAT-2 might consist of

21-ID-D:

We should consider changing 21-ID-D to a horizontally deflecting monochromator and a vertical goniometer along with a vertically deflecting undulator. These changes might be crucial to maintain beam stability for routine 1 micron sized crystals.

Our current mirror system is probably not good enough to fully utilize the MBA lattice. Consider replacement with state of the art (circa 2019) or replacement with compound refractive lenses or a combination of the two.

The currently proposed Pilatus 3 6M detector is probably the minimum entry level detector for handling single shot microcrystal data collection. Likely a faster detector with a smaller pixel size will be needed.

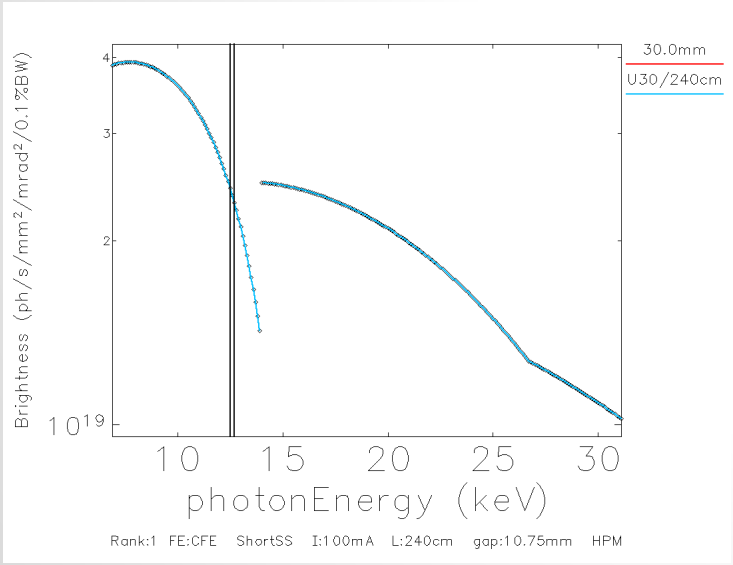
LCLS experiments are pioneering handling of micro crystal slurries. While there is a lot of work to be done before it becomes obvious which methods are best for us, it is clear that everything from crystal production to data analysis is very different from how things are done today.

What an LS-CAT-2 might consist of

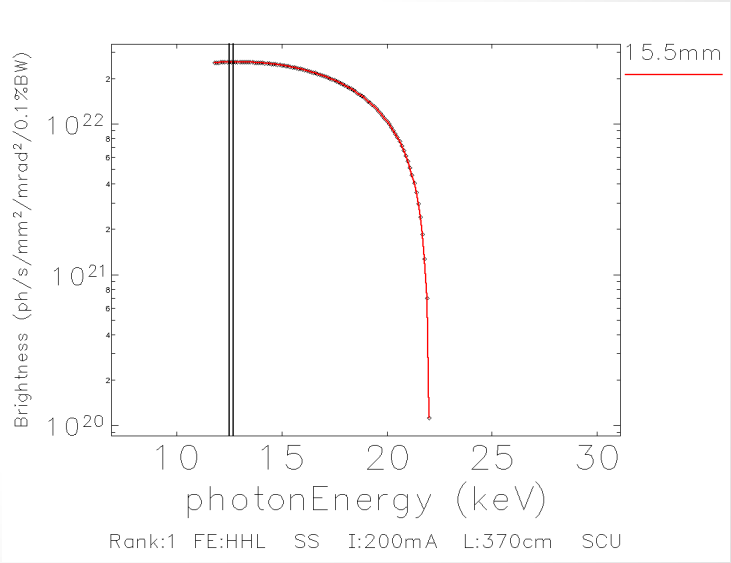
21-ID-F/G:

Since this branch of LS-CAT needs to work only in a tiny energy range we can configure the undulator for the Se edge.

New monos and updated lenses likely are needed to achieve maximum capabilities.



Current Undulator (U30S)



Potential Undulator (U15.5 SCU)

What an LS-CAT-2 might consist of

21-ID-E:

Could be adapted for a nano probe.

For all of LS-CAT:

All aspects of the beamlines need scrutiny, in particular the requirements for mirrors and lenses to take advantage of the increased brightness of the APS-U.

For members:

What capabilities would get you and your colleagues excited about an LS-CAT-2?

APS review of LS-CAT Nov. 4, 2014

Preparation for the APS review of LS-CAT is similar to what is needed to approach our administrations for extending the existing agreement and funding an LS-CAT-2:

- Science cases
- Publications and PDB deposits
- Size (people and dollars) of your research programs

We'll have room for 3 or so science talks. Please suggest a speaker or two.

We'll start asking for boilerplate info as 2014 progresses.

2013-3 LS-CAT Pilot Program: Data Collection and Structure Solution Workshop

- On Nov 4 & 5 LS-CAT hosted a user workshop
- 10 Participants
- 1st day covered
  - Data collection theory
  - Strategy theory
  - Processing programs
  - Remote Access
  - Data collection at the beamlines w/test xtals & student's projects
- 2nd day covered
  - General overview or structure determination pipelines
  - Data processing cutoffs & recent literature
  - Hands-on computer tutorials working with data at the beamlines

Future Workshop Plans

- Have more workshops on this scale, keep it small and more interactive
- Attend APS CCP4 workshop lectures
- Encourage members to apply to APS CCP4 workshop

- CALL for a list of student thesis work which used LS-CAT and meeting abstracts/presentations
  - Send the list to me: j-brunzelle@northwestern.edu

New Business?

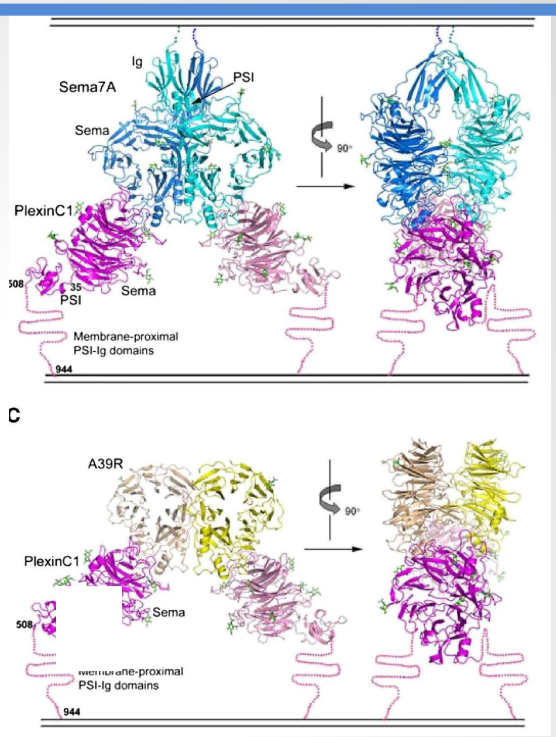
Impact on Northwestern researchers

Scientific Opportunity

Membrane proteins constitute one third of all human proteins and represent over 50% of all drug targets. Studies of membrane proteins are central to understanding fundamental processes in cells and also for drug development efforts.

Northwestern involvement

Several groups at Northwestern University (He, Lamb, Pinkett, and Rosenzweig) are actively studying important membrane proteins taking advantage of LS CAT for their studies.



Research from the He lab. FSM.

The Sema7A dimer (cyan and blue) bound to its receptor PlexinC1. The large 7-blade propellers of Sema7A and PlexinC1 interact side-to-side diagonally. A poxvirus-encoded soluble semaphorin, A39R, mimics Sema7A and efficiently engages the same PlexinC1 receptor with much smaller molecular mass. (From the he lab. Cell,142:749, 2010)

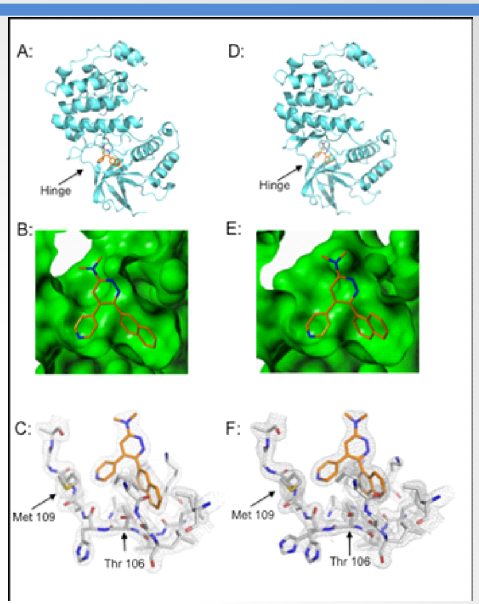
Impact on Northwestern researchers

Scientific Opportunity

Structure-based drug discovery is a growing area where synchrotron-based work promises to continue having a large impact. The ability to obtain information on important drug leads and to visualize potential compounds quickly and accurately will continue to grow rapidly.

Northwestern involvement

Groups at Northwestern University in several Departments have active drug-discovery efforts and their involvement in synchrotron-based research is growing rapidly. Already several groups use LS CAT routinely for their work.



Research from the Anderson,Tokars, Watterson lab. FSM.

Binding of drugs in the active site of human p38α MAPK: The structure of the kinase is bi-lobed with the drugs bound in the ATP region of the active site. The hinge region connects the two lobes and is denoted by the arrow. (A) X-ray crystal structure of 4F9W complexed with MW108. (B) Connolly surface generated from the coordinates of 4F9W. (C) Electron density (2mFo-DFc) map for active site region of 4F9W. Met109 is a key interaction point for the small molecule as well as the ATP substrate. Selectivity and affinity are mostly attained by combining this common point of protein interaction with a novel hydrophobic pocket nearby. (D) Structure of 4F9Y at 1.85 Å resolution. (E) Connolly surface generated from the coordinates of 4F9Y. (F) Electron density (2mFo-DFc) electron density of 4F9Y

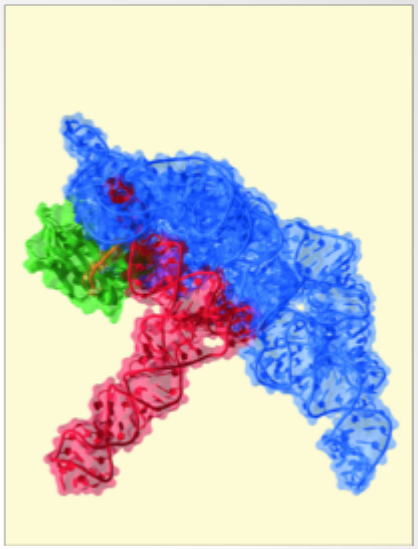
Impact on Northwestern researchers

Scientific Opportunity

Basic research on fundamental biological processes continues to provide new insights into the mechanism of the cell at the atomic level and to provide new information that has an impact on many aspects of human health research.

Northwestern involvement

A large number of groups at Northwestern University are actively involved in exploring fundamental aspects of the cell machinery, ranging from the role of RNA in the cell to the assembly and mechanism of complex molecular motors.



Research from the Mondragon lab.

Structure of bacterial RNase P holoenzyme in complex with tRNA. The RNA component is shown in blue, the protein component in green, tRNA in red, and the leader sequence in orange. (*Nature,* 468: 784, 2010)

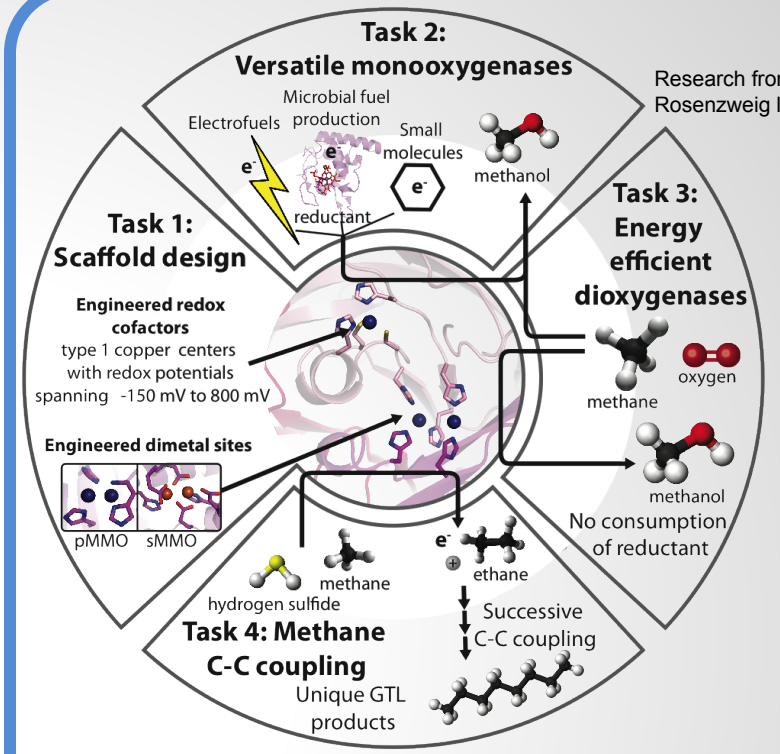
Impact on Northwestern researchers

Scientific Opportunity

Studies of proteins and other important biological macromolecules continues to grow and the interest of many Federal agencies on large-scale projects involving facile and rapid access to synchrotron radiation is sometimes key to the success of these efforts.

Northwestern involvement

Northwestern is involved in several large projects that rely heavily on access to synchrotron radiation. Current efforts on structural genomics (Anderson lab) depend on continuous use of LS CAT. More recent developments, including work on engineering of proteins for energy research (Rosenzweig lab) are also dependent on synchrotron-based research.



Engineering Multicopper Oxidases for Methane C-H Activation

Impact on Northwestern researchers

Scientific Opportunity

Precise determination of nanoparticle subcellular localization

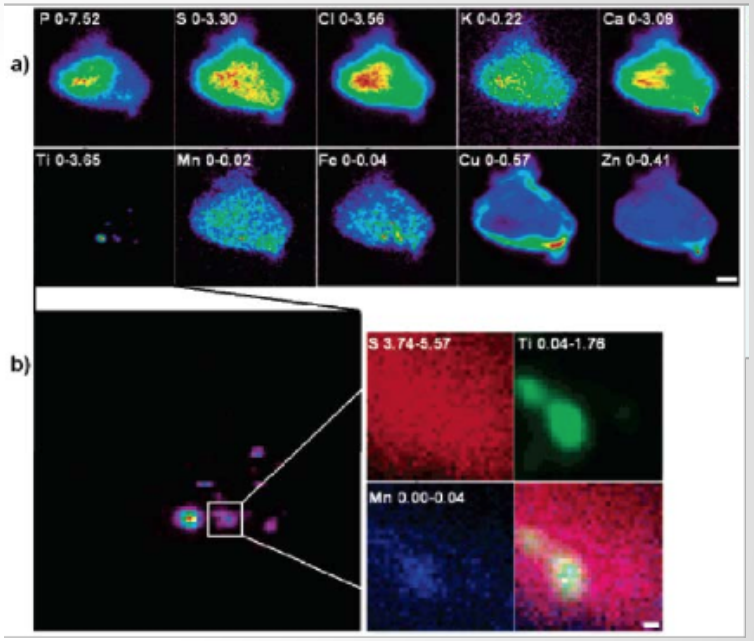
Northwestern involvement

Gayle Woloschak

Tom O'Halloran

and the interest is growing!

Research from the Woloschak lab



Paunesku et al. *Nano Lett.* 2007 Mar;7(3):596-601